Sahaj Bhadja

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Summary

Software engineer with 2+ years building cloud-native full-stack applications, secure REST APIs, microservices, data workflows, and AI-assisted products using React, TypeScript, Node.js, Python, SQL, Azure, Docker, and automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant **Azure SQL and Blob Storage** services for 500+ users, optimizing normalized schemas, indexed queries, and secure access patterns to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant **FastAPI workflows** for 1,000+ monthly AI jobs using explicit states, retries, validation, and failure isolation, tripling throughput while improving production reliability and debugging visibility.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered **Python REST APIs and React workflows** handling 10,000+ daily requests, integrating secure file ingestion, AWS RDS persistence, and AI-assisted grading while maintaining stable production performance.
- Designed concurrent batch processing for 1,000+ uploaded files using multithreading, automated validation, and fault-tolerant execution, reducing turnaround by 60% while strengthening testing and defect resolution under peak workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible Python data pipelines processing 100,000+ trajectory logs for autonomous-driving simulations, improving long-horizon planning by 18% through systematic testing, debugging, model evaluation, and secure experiment tracking.
- Architected a spatial-temporal encoder combining HD-map geometry and historical trajectories through multi-head attention, improving closed-loop simulation stability by 15% across complex, high-volume machine-learning data workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform **microservices** with Jenkins CI/CD and 20+ PyTest and Mockito suites, automating unit, integration, and regression testing across Windows, Linux, and macOS production environments.
- Optimized **REST API and ETL integrations** using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving distributed data-pipeline efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a **React and Node.js** platform with REST APIs, WebSockets, and eight modular services, supporting 2M+ synchronized records while translating specifications into extensible real-time workflows.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving code quality, documentation, and defect resolution across frequent production releases.

Skills

Full Stack & APIs: React, TypeScript, JavaScript, Node.js, Python, Java, SQL, HTML5, CSS3, FastAPI, Flask, REST APIs, JSON, WebSockets, microservices, service boundaries, object-oriented programming, SOLID, DRY, design patterns; **Data & AI:** PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB, Firebase, AWS RDS, ETL, data integration, data pipelines, analytics, machine learning, NLP, LLM orchestration, AI-assisted development, prompt engineering, human-in-the-loop workflows; **Cloud & Quality:** Azure, AWS, GCP, Blob Storage, S3, AWS Glue, Docker, Kubernetes, Jenkins, Git, CI/CD, PyTest, Mockito, unit testing, integration testing, regression testing, debugging, code review, performance optimization, monitoring, Agile/Scrum, security, HIPAA-aware development

Projects

HMIS Live | *TypeScript, MERN, ABDM APIs, Sentry, Twilio*
- Built a **human-services healthcare platform** integrating consent-based APIs, appointment workflows, Twilio alerts, and Sentry monitoring, reducing administrative workload by 30% across secure patient and clinician operations.

Big-Brain Live | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built an AI data platform for 200+ users with secure ingestion, vector indexing, semantic retrieval, and collaborative workflows, maintaining 95% query accuracy across 50,000+ governed documents.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, REST APIs*
- Built an AI-assisted Java and TypeScript database workspace supporting 10+ systems, accelerating SQL authoring, schema exploration, validation, and relational data management through a unified cross-platform interface.